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Course code: DSA0602

Course name: Data Handling and Visualization

LAB PROGRAMS (8-12)

SET 8 EMPLOYEE PERFORMANCE ANALYSIS

Dataset: Employee Performance

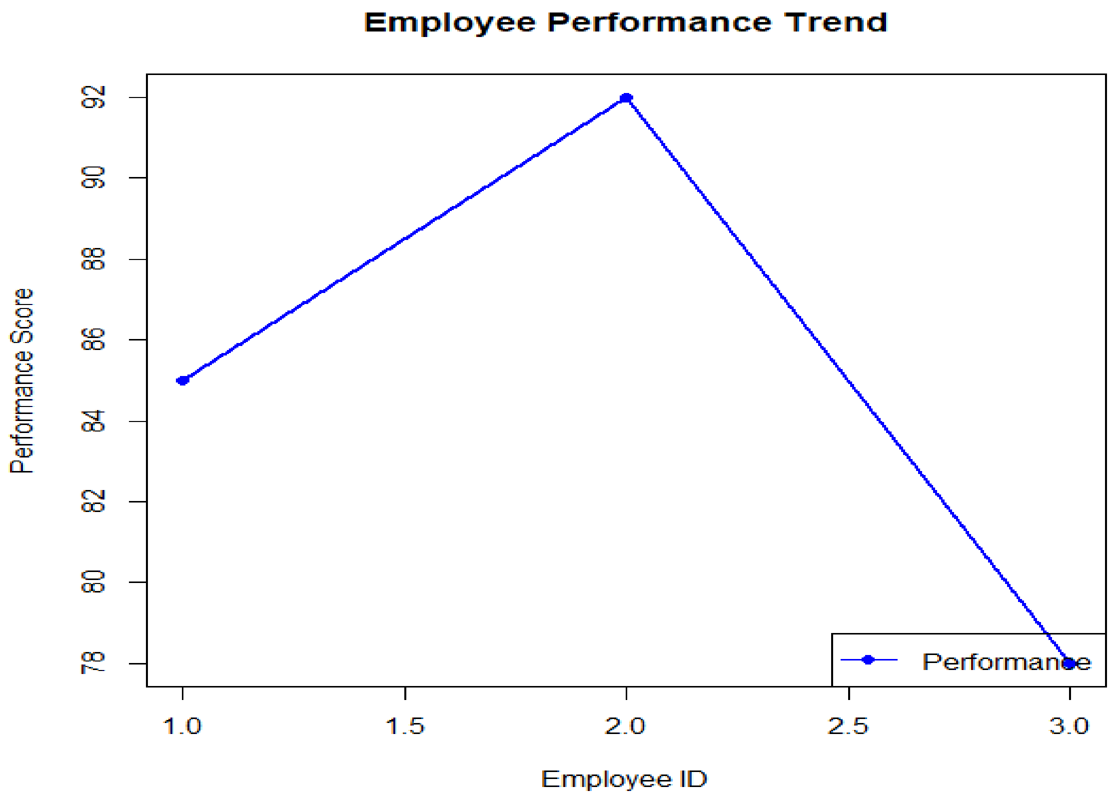
```
employee <- data.frame(
  EmployeeID = c(1,2,3),
  Department = c("Sales","HR","Marketing"),
  YearsService = c(5,3,7),
  PerformanceScore = c(85,92,78)
)

employee
```

```
> employee
  EmployeeID Department YearsService PerformanceScore
1          1      Sales            5               85
2          2         HR            3               92
3          3  Marketing            7               78
> 
```

1. **Create a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.**

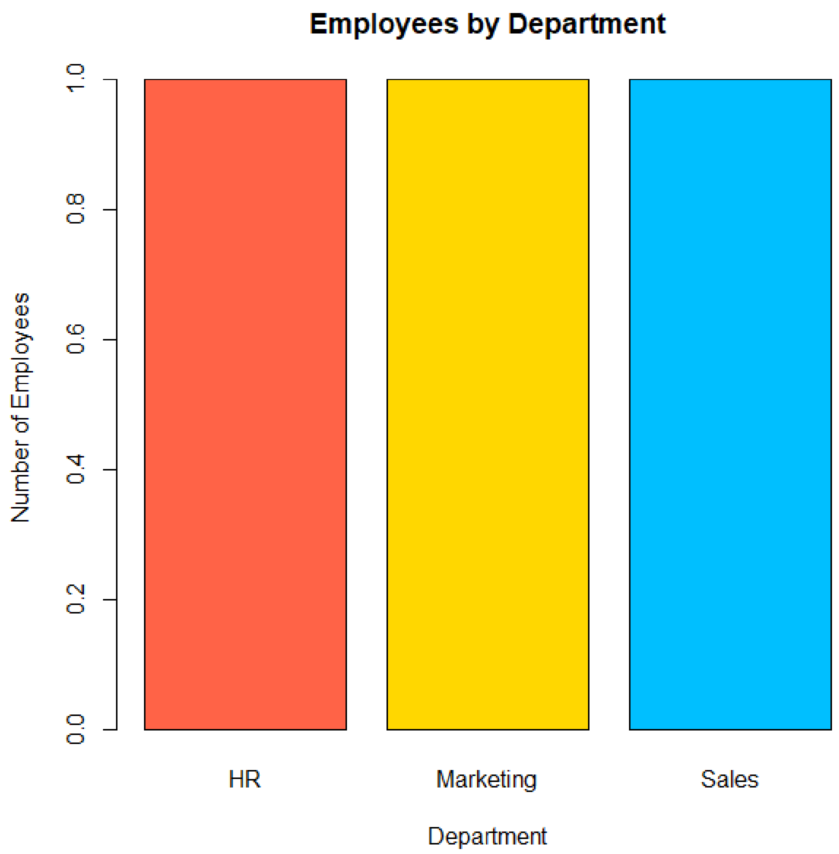
```
employee <- data.frame(
  EmployeeID = c(1,2,3),
  Department = c("Sales","HR","Marketing"),
  YearsService = c(5,3,7),
  PerformanceScore = c(85,92,78)
)

plot(employee$EmployeeID,
     employee$PerformanceScore,
     type="o",
     pch=16,
     col="blue",
     lwd=2,
```

```
    xlab="Employee ID",
    ylab="Performance Score",
    main="Employee Performance Trend")

legend("bottomright",
      legend="Performance",
      col="blue",
      pch=16,
      lty=1)
```



2. **Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.**

```
dept <- table(employee$Department)

barplot(dept,
      col=c("tomato","gold","deepskyblue"),
      border="black",
      xlab="Department",
      ylab="Number of Employees",
      main="Employees by Department")
```



3. Build a table to display the performance data for each employee. Label the table elements.

```
employee <- data.frame(
  EmployeeID = c(1,2,3),
  Department = c("Sales","HR","Marketing"),
  YearsService = c(5,3,7),
  PerformanceScore = c(85,92,78)
)

employee$Grade <- ifelse(employee$PerformanceScore >= 90,"A",
            ifelse(employee$PerformanceScore >= 80,"B","C"))

print(employee)
```

```
print(employee)
  EmployeeID Department YearsService PerformanceScore Grade
1          1      Sales            5               85     B
2          2         HR            3               92     A
3          3  Marketing            7               78     C
```

4. **Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive exploration of employee performance data.**

```
library(shiny)
employee <- data.frame(
  EmployeeID = c(1,2,3),
  Department = c("Sales","HR","Marketing"),
  YearsService = c(5,3,7),
  PerformanceScore = c(85,92,78)
)
employee$Grade <- ifelse(employee$PerformanceScore >= 90,"A",
               ifelse(employee$PerformanceScore >= 80,"B","C"))
ui <- fluidPage(
titlePanel("Employee Performance Dashboard"),
sidebarLayout(
sidebarPanel
selectInput("view",
          "Select View",
          choices=c("Line Chart",
                  "Bar Chart",
                  "Employee Table"))

),

mainPanel(

plotOutput("plot"),

tableOutput("table")

)

)

)

server <- function(input,output){

output$plot <- renderPlot({

if(input$view=="Line Chart"){
```

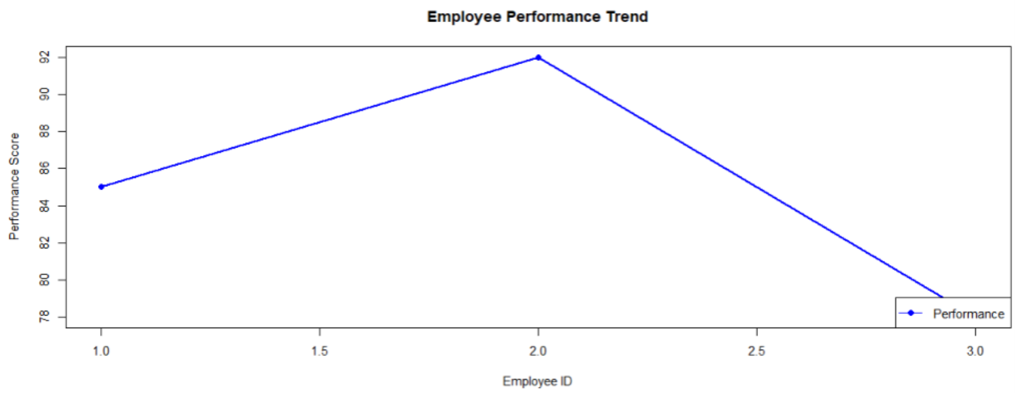
```
plot(employee$EmployeeID,
    employee$PerformanceScore,
    type="o",
    pch=16,
    col="blue",
    lwd=2,
    xlab="Employee ID",
    ylab="Performance Score",
    main="Employee Performance Trend")

legend("bottomright",
      legend="Performance",
      col="blue",
      pch=16,
      lty=1)

}

else if(input$view=="Bar Chart"){

dept <- table(employee$Department)

barplot(dept,
       col=c("tomato","gold","deepskyblue"),
       border="black",
       xlab="Department",
       ylab="Number of Employees",
       main="Employees by Department")

}

})

output$table <- renderTable({

if(input$view=="Employee Table"){

employee

}
```

```
})

}

shinyApp(ui=ui,server=server)
```

Employee Performance Dashboard

Select View

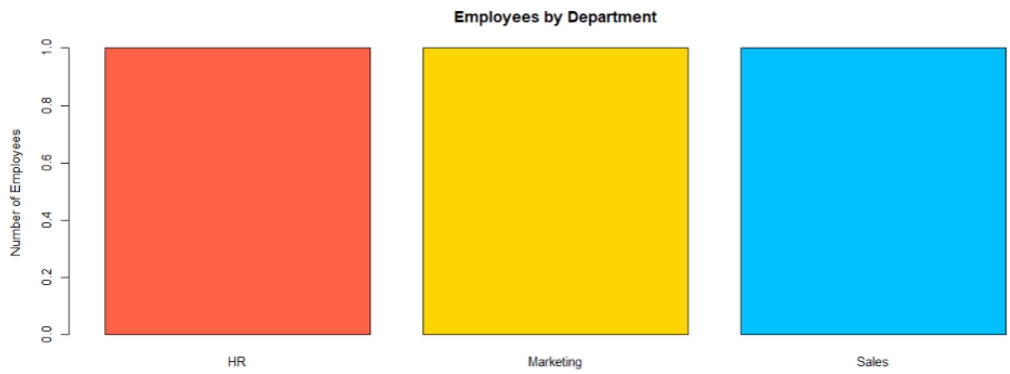
Line Chart



Employee Performance Dashboard

Select View

Bar Chart



Employee Performance Dashboard

Select View

Employee Table

EmployeeID	Department	YearsService	PerformanceScore	Grade
1.00	Sales	5.00	85.00	B
2.00	HR	3.00	92.00	A
3.00	Marketing	7.00	78.00	C

SET 9

Dataset: Online_Learning_Activity.csv

```
data <- data.frame(
  Student_ID = c("L01","L02","L03","L04","L05","L06"),
  Gender = c("Male","Female","Male","Female","Male","Female"),
  Age = c(20,22,19,21,23,20),
  Course = c("R","R","SQL","R","R","SQL"),
  Study_Time = c(3.5,4.2,2.0,5.0,2.5,4.0),
  Videos_Watched = c(12,15,8,18,9,14),
  Quiz_Score = c(78,85,65,92,70,88),
  Login_Date = c("2025-01-05","2025-01-05","2025-02-08",
              "2025-02-08","2025-03-12","2025-03-12")
)
```

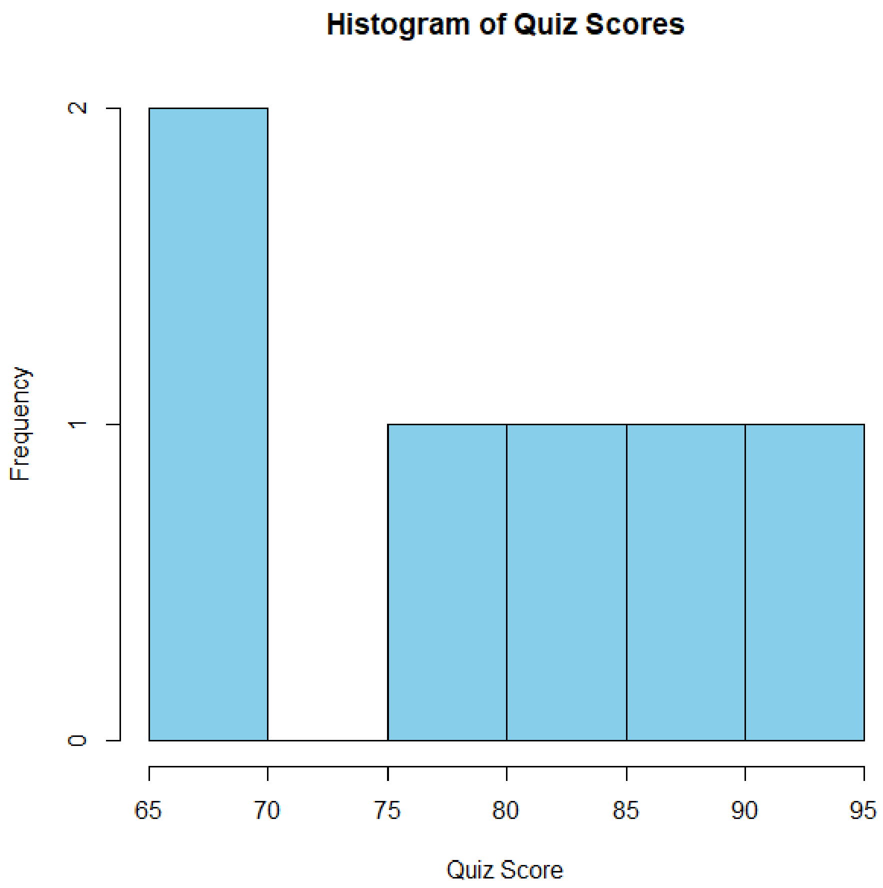
Data

```
> data
  Student_ID Gender Age Course Study_Time Videos_Watched Quiz_Score Login_Date
1        L01   Male  20      R        3.5             12         78 2025-01-05
2        L02 Female  22      R        4.2             15         85 2025-01-05
3        L03   Male  19    SQL        2.0              8         65 2025-02-08
4        L04 Female  21      R        5.0             18         92 2025-02-08
5        L05   Male  23      R        2.5              9         70 2025-03-12
6        L06 Female  20    SQL        4.0             14         88 2025-03-12
> 
```

1. **Import the dataset in R. Plot a histogram for Quiz_Score. Draw a boxplot of Quiz_Score by Course. Add title and labels. Interpret student performance.**

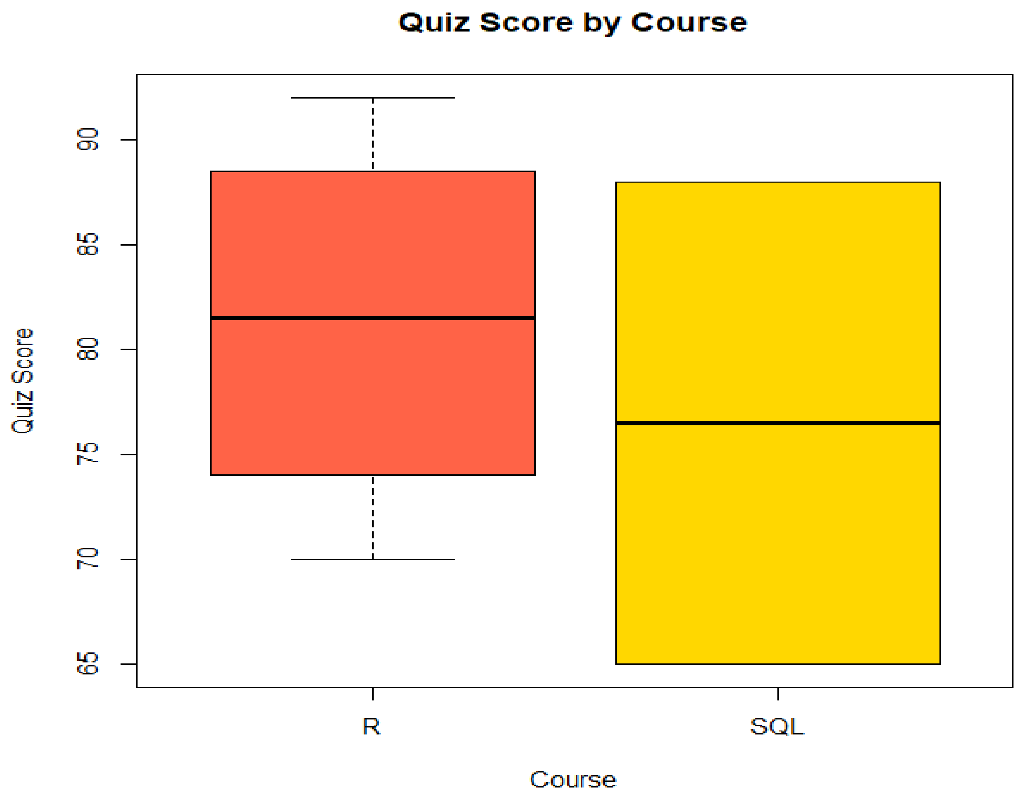
Histogrim of quiz scores:

```
hist(data$Quiz_Score,
    col="skyblue",
    border="black",
    main="Histogram of Quiz Scores",
    xlab="Quiz Score",
    ylab="Frequency")
```



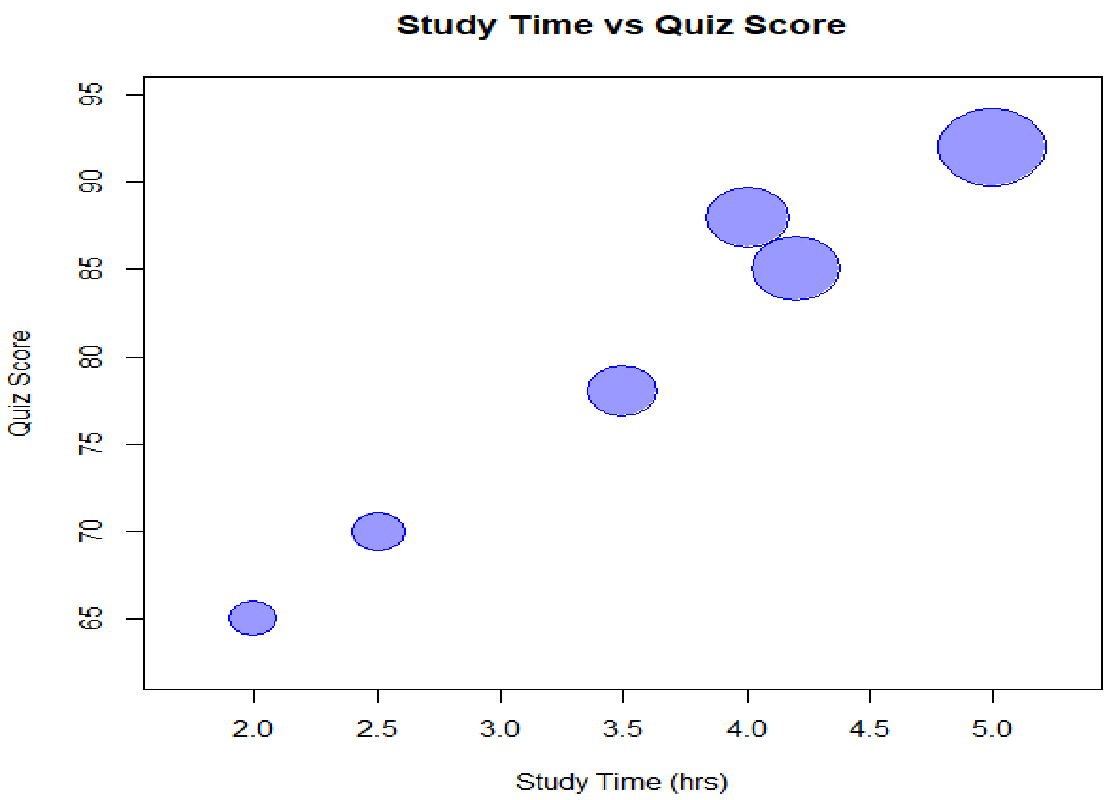
Boxplot of quiz scores:

```
boxplot(Quiz_Score~Course,
    data=data,
    col=c("tomato","gold"),
    main="Quiz Score by Course",
    xlab="Course",
    ylab="Quiz Score")
```



2. **Create a scatter plot using R programming between Study_Time and Quiz_Score. Represent Videos_Watched using bubble size. Apply transparency. Explain the learning pattern.**

```
symbols(data$Study_Time,
      data$Quiz_Score,
      circles=data$Videos_Watched,
      inches=0.3,
      bg=rgb(0,0,1,0.4),
      fg="blue",
      xlab="Study Time (hrs)",
      ylab="Quiz Score",
      main="Study Time vs Quiz Score")
```



3. **Using R programming, convert Login_Date into Date format. Calculate average quiz score per month. Plot a line chart. Apply moving average smoothing. Identify the trend.**

Convert Date and Calculate Monthly Average:

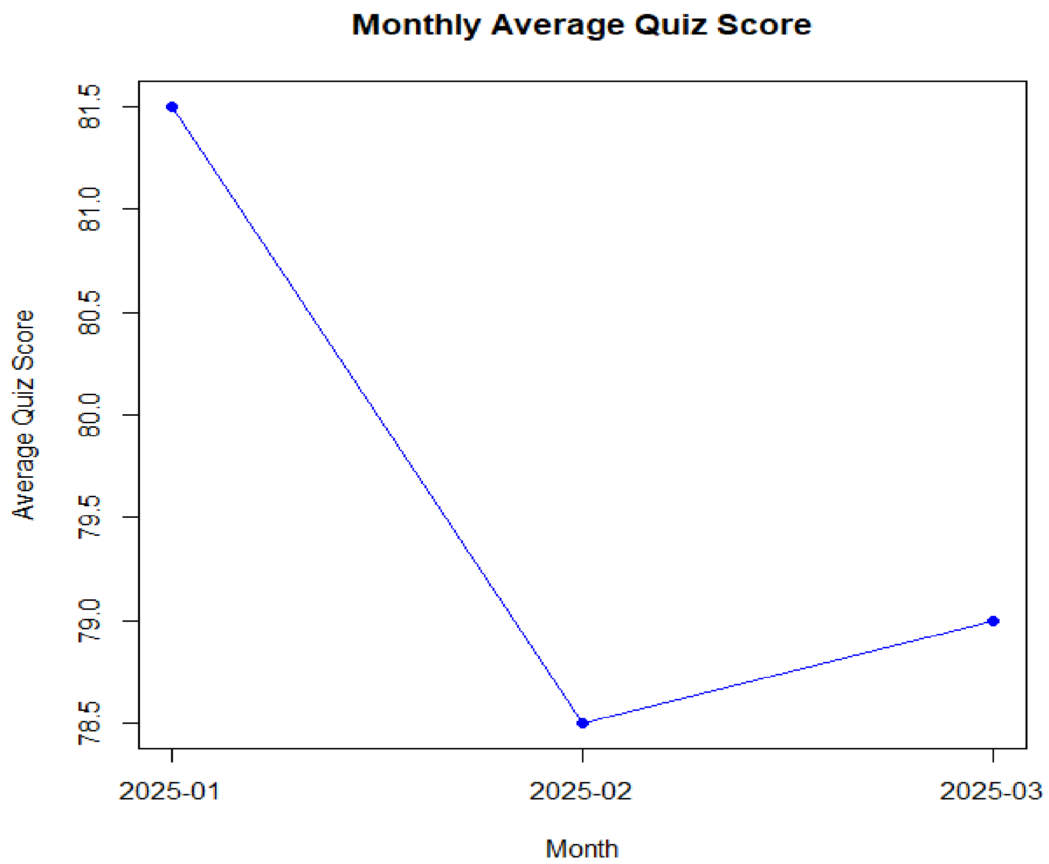
```
data$Login_Date <- as.Date(data$Login_Date)
data$Month <- format(data$Login_Date,"%Y-%m")
avg <- aggregate(Quiz_Score~Month,
          data=data,
          mean)
print(avg)
```

```
> print(avg)
    Month Quiz_Score
1 2025-01       81.5
2 2025-02       78.5
3 2025-03       79.0
>
```

Line Chart

```
plot(avg$Quiz_Score,
   type="o",
```

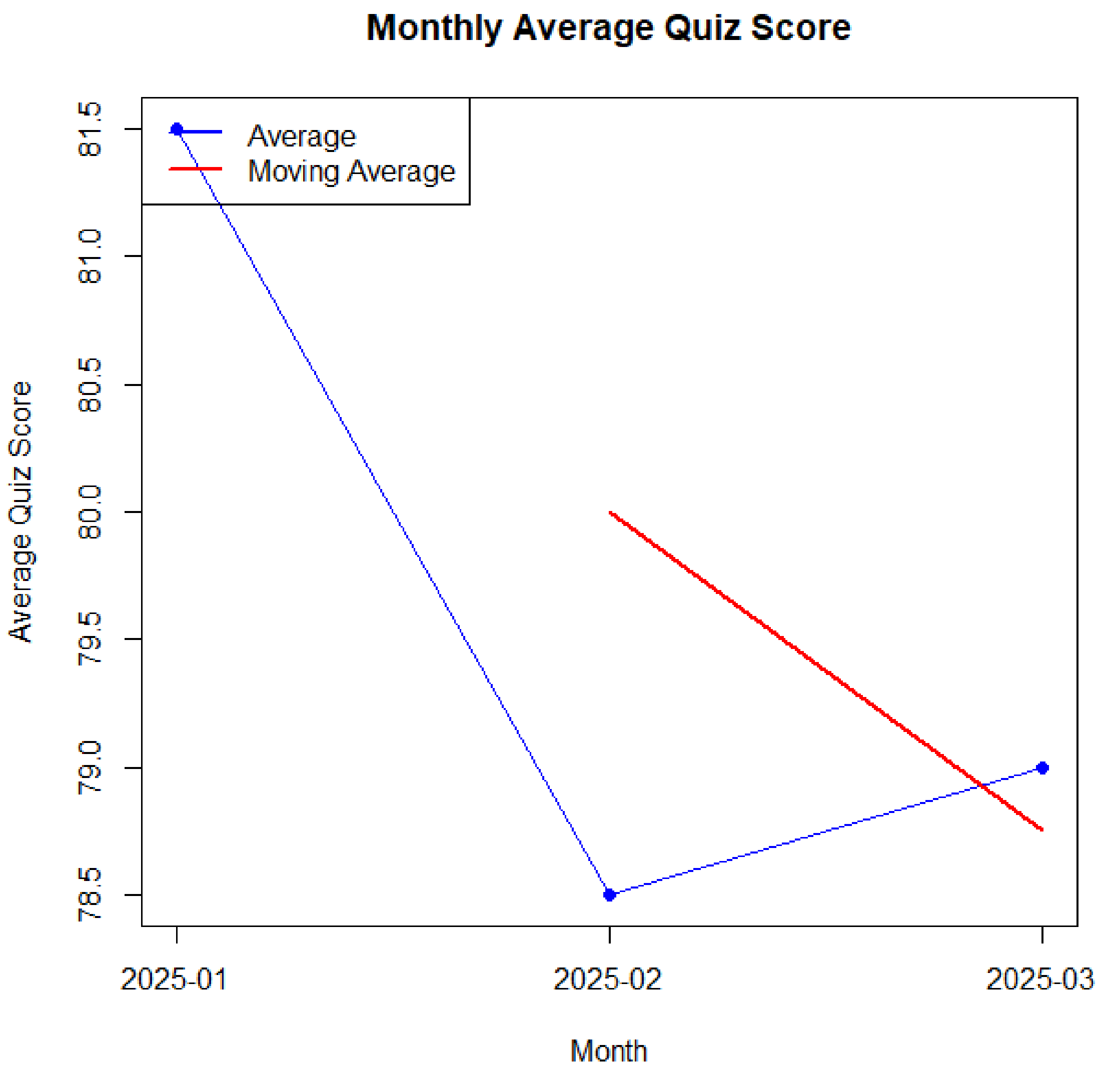
```
    pch=16,
    col="blue",
    xaxt="n",
    xlab="Month",
    ylab="Average Quiz Score",
    main="Monthly Average Quiz Score")

axis(1,
    at=1:nrow(avg),
    labels=avg$Month)
```



Moving Average:

```
ma <- stats::filter(avg$Quiz_Score,
                rep(1/2,2),
                sides=1)

lines(ma,
     col="red",
     lwd=2)
```

```
legend("topleft",
    legend=c("Average","Moving Average"),
    col=c("blue","red"),
    lty=1,
    lwd=2)
```



4. Create a bar chart showing average quiz score by Course. Create a scatter plot (Study Time vs Quiz Score). Add filter for Gender and Course. Design a mini interactive dashboard.

```
library(shiny)

data <- data.frame(
 Student_ID=c("L01","L02","L03","L04","L05","L06"),
 Gender=c("Male","Female","Male","Female","Male","Female"),
 Age=c(20,22,19,21,23,20),
 Course=c("R","R","SQL","R","R","SQL"),
```

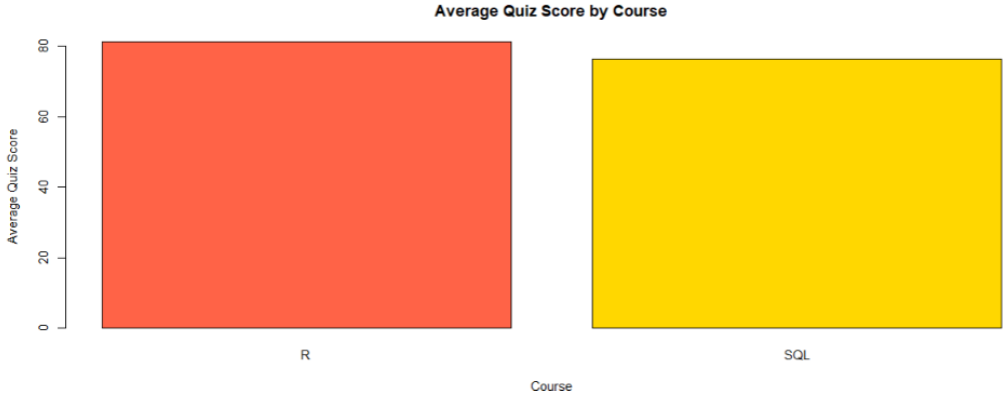
```
  Study_Time=c(3.5,4.2,2.0,5.0,2.5,4.0),
  Videos_Watched=c(12,15,8,18,9,14),
  Quiz_Score=c(78,85,65,92,70,88)
)

ui <- fluidPage(

titlePanel("Online Learning Activity Dashboard"),

sidebarLayout(

sidebarPanel(

selectInput("view",
        "Select View",
        choices=c("Bar Chart",
              "Scatter Plot",
              "Student Table"))

),

mainPanel(

plotOutput("plot"),
tableOutput("table")

)

)

)

server <- function(input,output){

output$plot <- renderPlot({

if(input$view=="Bar Chart"){

avg <- aggregate(Quiz_Score~Course,
          data=data,
          mean)
```

```
barplot(avg$Quiz_Score,
        names.arg=avg$Course,
        col=c("tomato","gold"),
        border="black",
        xlab="Course",
        ylab="Average Quiz Score",
        main="Average Quiz Score by Course")

}

else if(input$view=="Scatter Plot"){

symbols(data$Study_Time,
        data$Quiz_Score,
        circles=data$Videos_Watched,
        inches=0.3,
        bg=rgb(0,0,1,0.4),
        fg="blue",
        xlab="Study Time (hrs)",
        ylab="Quiz Score",
        main="Study Time vs Quiz Score")

}

})

output$table <- renderTable({

if(input$view=="Student Table"){

data

}

})

}

shinyApp(ui=ui,server=server)
```

Online Learning Activity Dashboard

Select View

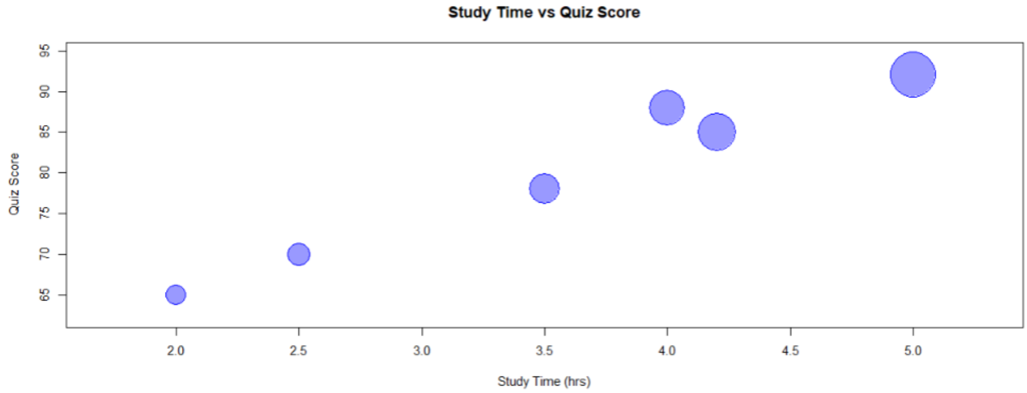
Bar Chart



Online Learning Activity Dashboard

Select View

Scatter Plot



Online Learning Activity Dashboard

Select View

Student Table

Student_ID	Gender	Age	Course	Study_Time	Videos_Watched	Quiz_Score
L01	Male	20.00	R	3.50	12.00	78.00
L02	Female	22.00	R	4.20	15.00	85.00
L03	Male	19.00	SQL	2.00	8.00	65.00
L04	Female	21.00	R	5.00	18.00	92.00
L05	Male	23.00	R	2.50	9.00	70.00
L06	Female	20.00	SQL	4.00	14.00	88.00

SET 10 SURVEY RESPONSES ANALYSIS

Dataset: Survey Responses

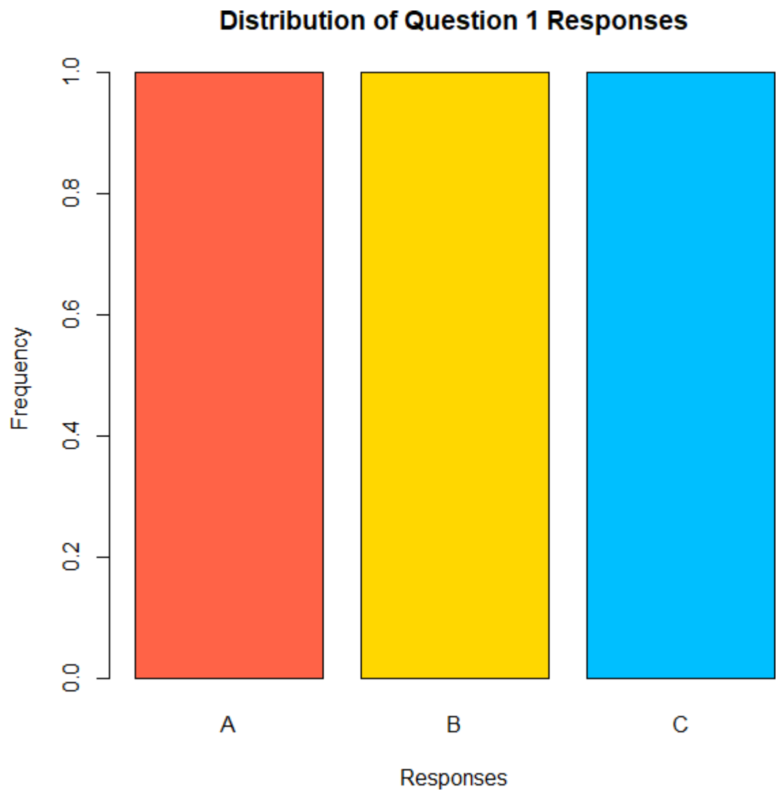
```
survey <- data.frame(
SurveyID = c(1,2,3),
  Question1 = c("A","B","C"),
  Question2 = c("B","A","A"),
  Question3 = c("C","D","B")
)
```

Survey

```
> survey
  SurveyID Question1 Question2 Question3
1        1         A         B         C
2        2         B         A         D
3        3         C         A         B
> 
```

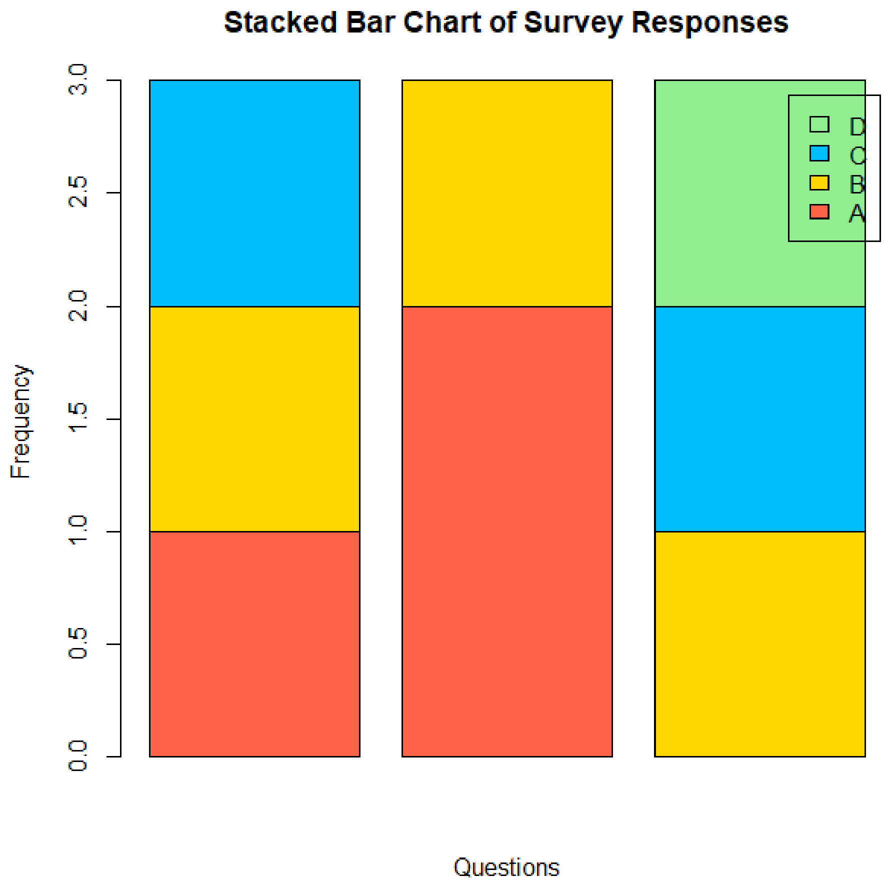
1. **Create a grouped bar chart to visualize the distribution of answers for Question 1. Label the chart elements.**

```
survey <- data.frame(
  SurveyID = c(1,2,3),
  Question1 = c("A","B","C"),
  Question2 = c("B","A","A"),
  Question3 = c("C","D","B")
)

q1 <- table(survey$Question1)

barplot(q1,
        beside=TRUE,
        col=c("tomato","gold","deepskyblue"),
        border="black",
        main="Distribution of Question 1 Responses",
        xlab="Responses",
        ylab="Frequency")
```



2. Generate a stacked bar chart to represent the overall distribution of responses for all three questions.

```
survey <- data.frame(
  SurveyID = c(1,2,3),
  Question1 = c("A","B","C"),
  Question2 = c("B","A","A"),
  Question3 = c("C","D","B")
)

response_table <- rbind(
  table(factor(survey$Question1, levels=c("A","B","C","D"))),
  table(factor(survey$Question2, levels=c("A","B","C","D"))),
  table(factor(survey$Question3, levels=c("A","B","C","D")))
)

barplot(t(response_table),
        beside=FALSE,
        col=c("tomato","gold","deepskyblue","lightgreen"),
        legend.text=c("A","B","C","D"),
        xlab="Questions",
        ylab="Frequency",
        main="Stacked Bar Chart of Survey Responses")
```



3. **Build a table to show the survey response data for each survey. Label the table elements.**

```
survey <- data.frame(
  SurveyID = c(1,2,3),
  Question1 = c("A","B","C"),
  Question2 = c("B","A","A"),
  Question3 = c("C","D","B")
)

survey$TotalResponses <- 3

print(survey)
```

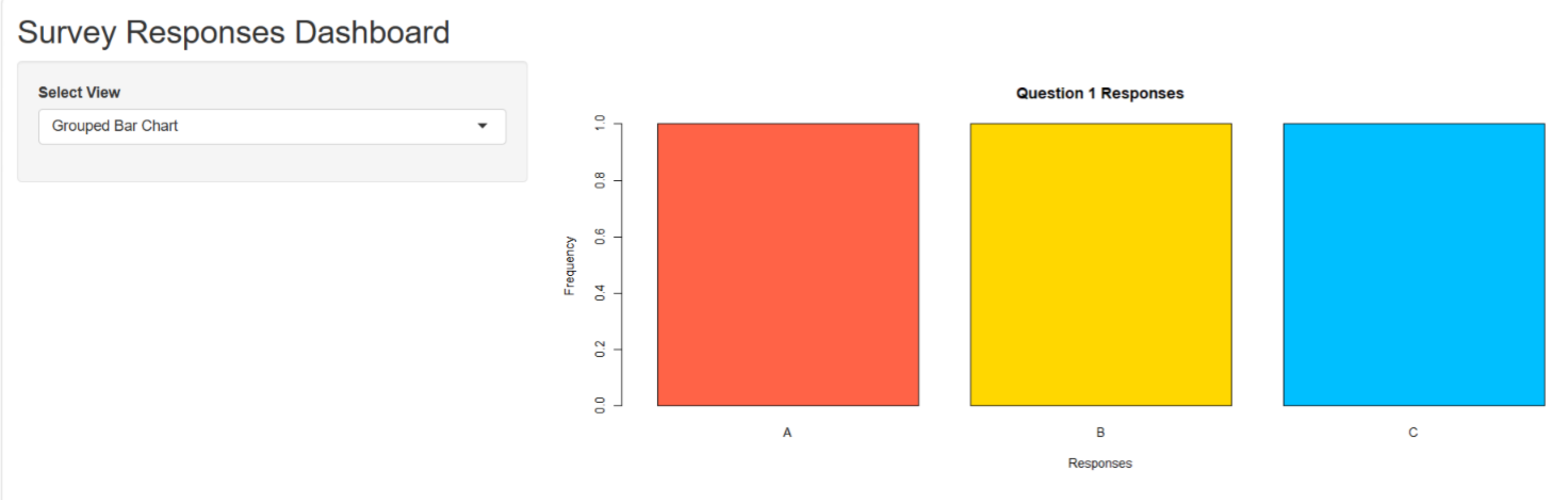
```
> print(survey)
  SurveyID Question1 Question2 Question3 TotalResponses
1        1         A         B         C              3
2        2         B         A         D              3
3        3         C         A         B              3
>
```

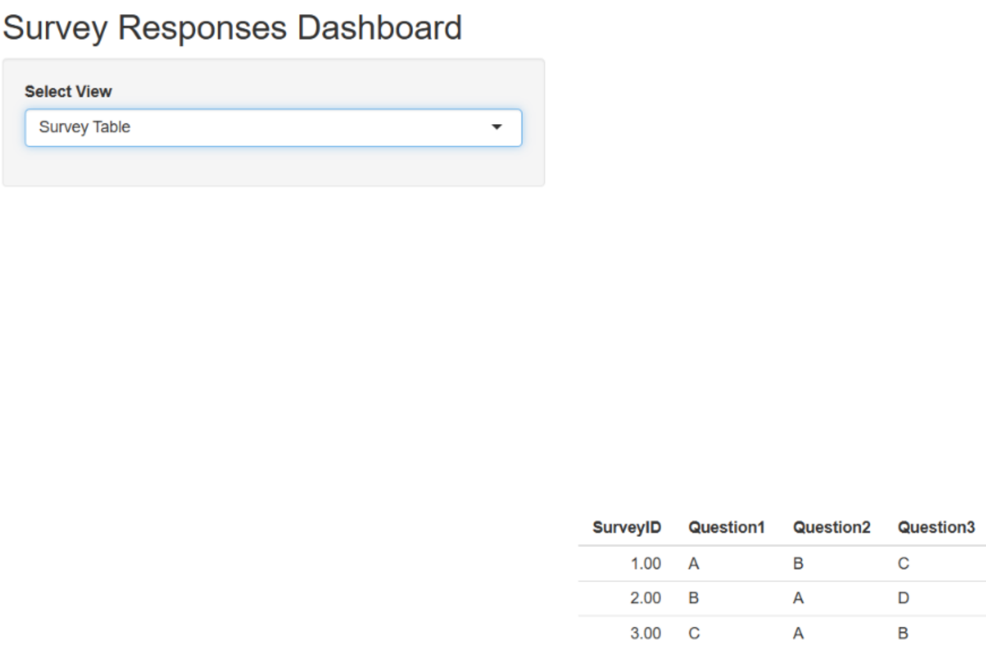
4. **Develop a Tableau dashboard combining the grouped bar chart, stacked bar chart, and the table for interactive exploration of survey responses.**

```
library(shiny)


survey <- data.frame(
  SurveyID = c(1,2,3),
  Question1 = c("A","B","C"),
  Question2 = c("B","A","A"),
  Question3 = c("C","D","B")
)

ui <- fluidPage(

titlePanel("Survey Responses Dashboard"),

sidebarLayout(

sidebarPanel(

selectInput("view",
            "Select View",
            choices=c("Grouped Bar Chart",
                      "Stacked Bar Chart",
                      "Survey Table"))

),

mainPanel(

plotOutput("plot"),
tableOutput("table")

)

)

)

server <- function(input,output){
```

```
output$plot <- renderPlot({

if(input$view=="Grouped Bar Chart"){

q1 <- table(survey$Question1)

barplot(q1,
        col=c("tomato","gold","deepskyblue"),
        border="black",
        main="Question 1 Responses",
        xlab="Responses",
        ylab="Frequency")

}

else if(input$view=="Stacked Bar Chart"){

response_table <- rbind(
table(factor(survey$Question1,levels=c("A","B","C","D"))),
table(factor(survey$Question2,levels=c("A","B","C","D"))),
table(factor(survey$Question3,levels=c("A","B","C","D")))
)

barplot(t(response_table),
        beside=FALSE,
        col=c("tomato","gold","deepskyblue","lightgreen"),
        legend.text=c("A","B","C","D"),
        xlab="Questions",
        ylab="Frequency",
        main="Survey Responses")

}

})

output$table <- renderTable({

if(input$view=="Survey Table"){

survey
```

```
}

})

}

shinyApp(ui=ui,server=server)
```





SurveyID	Question1	Question2	Question3
1.00	A	B	C
2.00	B	A	D
3.00	C	A	B

SET 11: PRODUCT CATEGORY ANALYSIS

Dataset: Product Categories

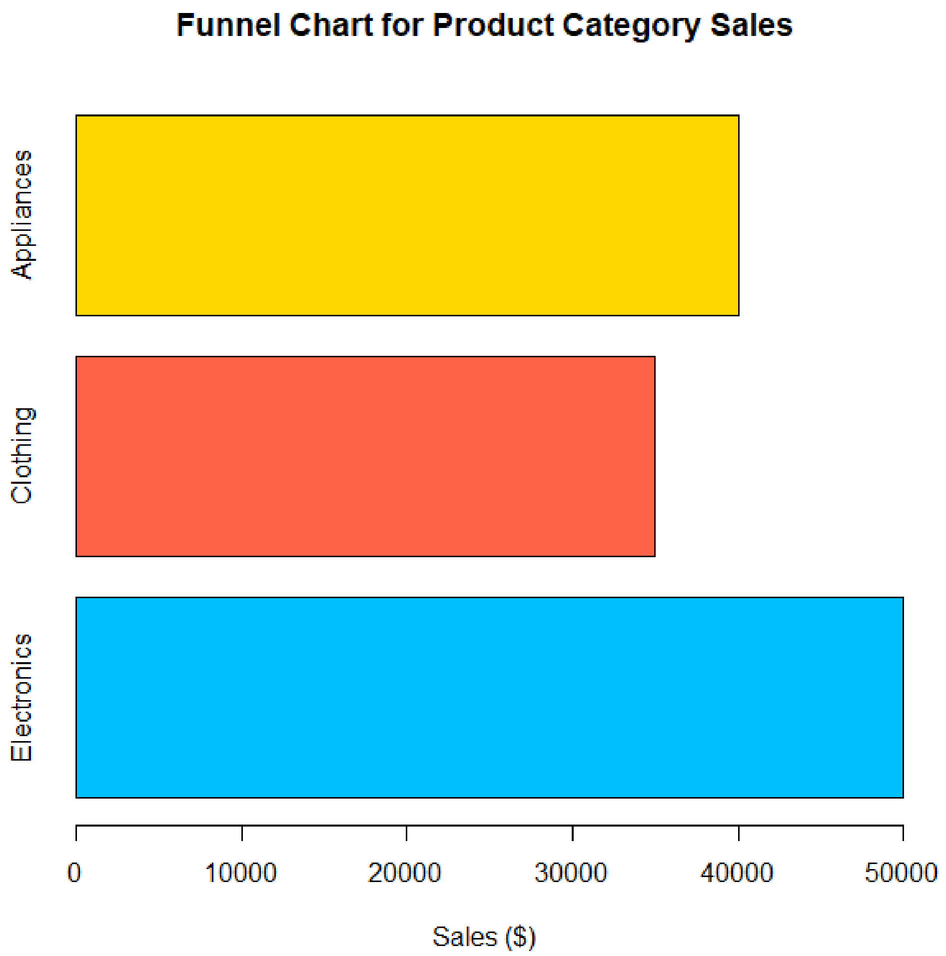
```
product <- data.frame(
  Category = c("Electronics","Clothing","Appliances"),
  Sales = c(50000,35000,40000)
)
```

Product

```
> product
     Category Sales
1 Electronics 50000
2    Clothing 35000
3  Appliances 40000
> 
```

1. Generate a funnel chart to analyze the sales conversion process for each product category. Label the stages and title the chart.

```
product <- data.frame(
  Category = c("Electronics","Clothing","Appliances"),
  Sales = c(50000,35000,40000)
)

barplot(product$Sales,
      horiz=TRUE,
      names.arg=product$Category,
      col=c("deepskyblue","tomato","gold"),
      border="black",
      xlab="Sales ($)",
      main="Funnel Chart for Product Category Sales")
```



2. Build a table to display the sales data for each product category. Label the table elements.

```
product <- data.frame(
  Category=c("Electronics","Clothing","Appliances"),
  Sales=c(50000,35000,40000)
)

product$Sales_in_Lakhs <- product$Sales/100000

print(product)
```

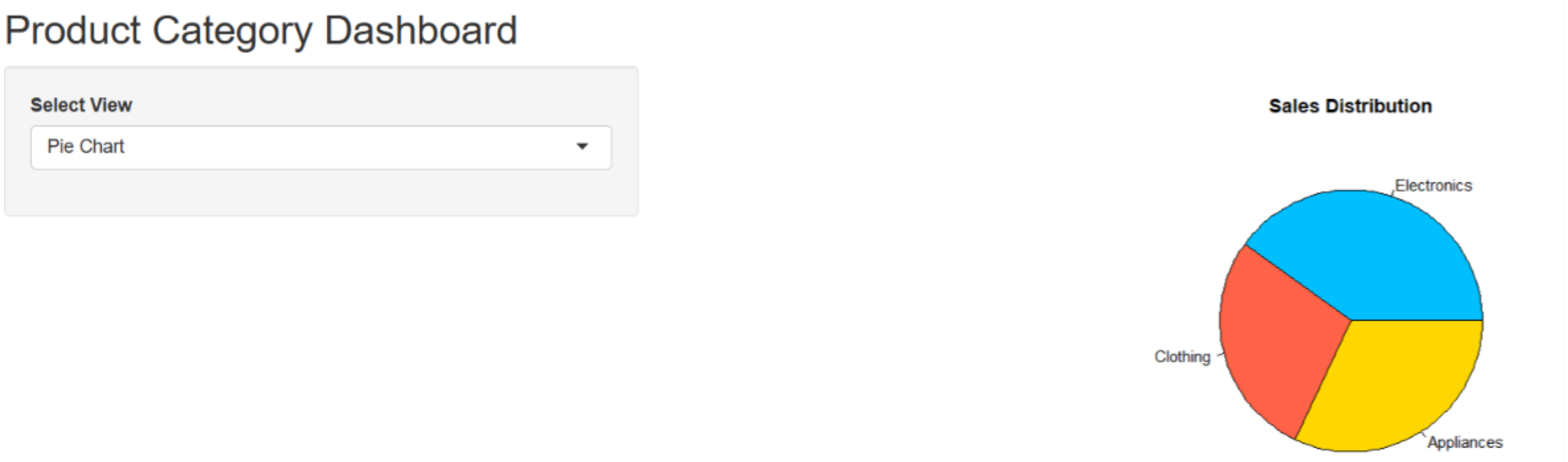
```
> print(product)
     Category Sales Sales_in_Lakhs
1 Electronics 50000           0.50
2    Clothing 35000           0.35
3  Appliances 40000           0.40
> 
```

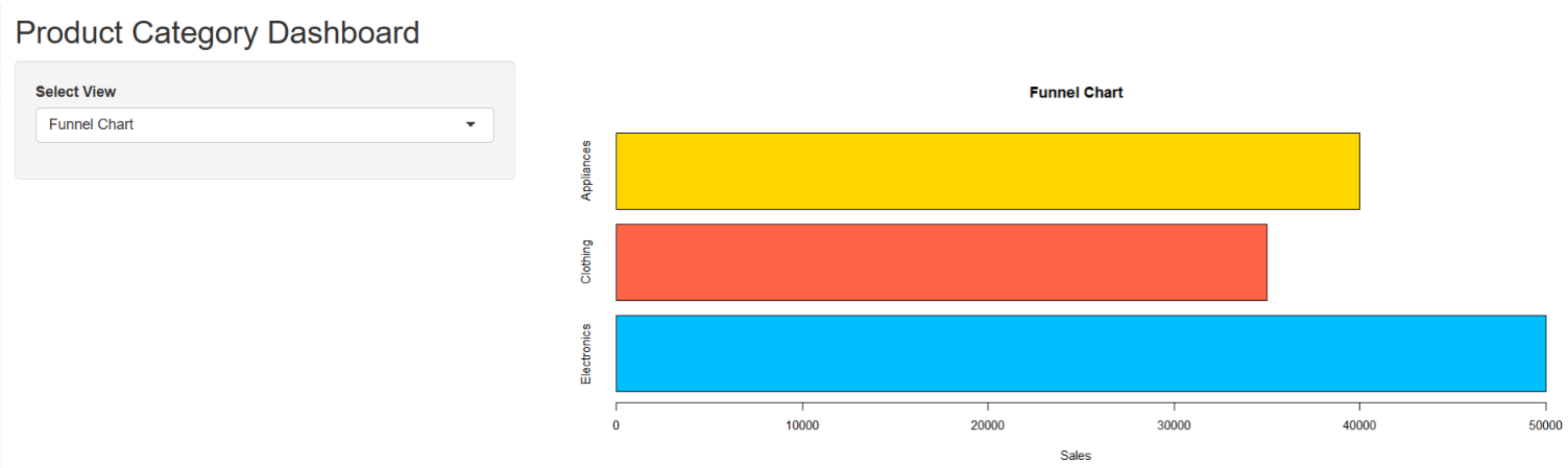
3. Develop a Tableau dashboard combining the pie chart, funnel chart, and the table for interactive exploration of product category data.

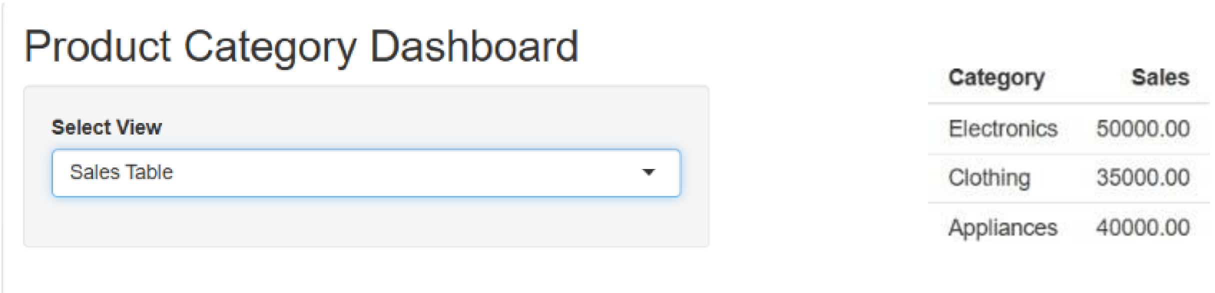
```
library(shiny)

product <- data.frame(
  Category=c("Electronics","Clothing","Appliances"),
  Sales=c(50000,35000,40000)
)

ui <- fluidPage(

titlePanel("Product Category Dashboard"),

sidebarLayout(

sidebarPanel(

selectInput("view",
        "Select View",
        choices=c("Pie Chart",
              "Funnel Chart",
              "Sales Table"))

),

mainPanel(

plotOutput("plot"),

tableOutput("table")

)

)
```

```
)

server <- function(input,output){

output$plot <- renderPlot({

if(input$view=="Pie Chart"){

pie(product$Sales,
   labels=product$Category,
   col=c("deepskyblue","tomato","gold"),
   main="Sales Distribution")

}

else if(input$view=="Funnel Chart"){

barplot(product$Sales,
        horiz=TRUE,
        names.arg=product$Category,
        col=c("deepskyblue","tomato","gold"),
        border="black",
        xlab="Sales",
        main="Funnel Chart")

}

})

output$table <- renderTable({

if(input$view=="Sales  Table"){

product

}

})

}
```

```
shinyApp(ui,server)
```







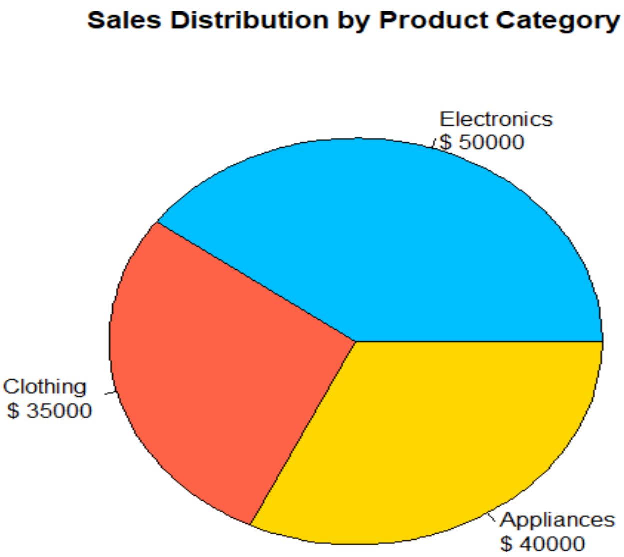
4. Create a pie chart to represent the distribution of sales across product categories. Include labels.

```
product <- data.frame(
  Category=c("Electronics","Clothing","Appliances"),
  Sales=c(50000,35000,40000)
```

)

pie(product$Sales,
labels=paste(product$Category,"\n$",product$Sales),
col=c("deepskyblue","tomato","gold"),

main="Sales Distribution by Product Category")



SET 12

Dataset: Website Traffic

traffic <- data.frame(

Date=c("2023-01-01","2023-01-02","2023-01-03"),

PageViews=c(1500,1600,1400),

CTR=c(2.3,2.7,2.0)

)

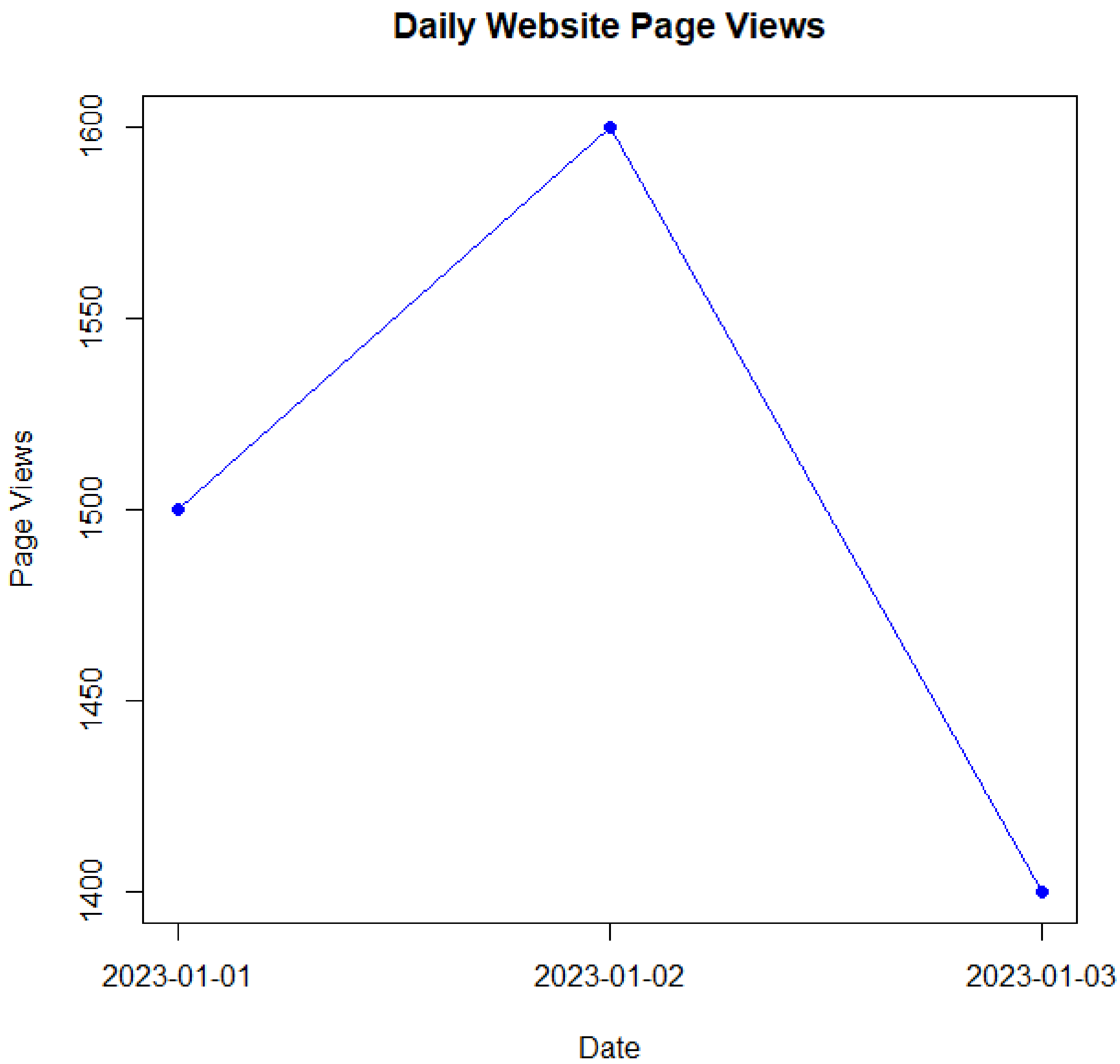
Traffic

```
> traffic
        Date PageViews CTR
1 2023-01-01      1500 2.3
2 2023-01-02      1600 2.7
3 2023-01-03      1400 2.0
>
```

1. **Create a map chart to visualize the distribution of cities on a geographic map. Label the map elements.**

```
plot(traffic$PageViews,
    type="o",
    pch=16,
    col="blue",
    xaxt="n",
    xlab="Date",
    ylab="Page Views",
    main="Daily Website Page Views")

axis(1,
    at=1:3,
```

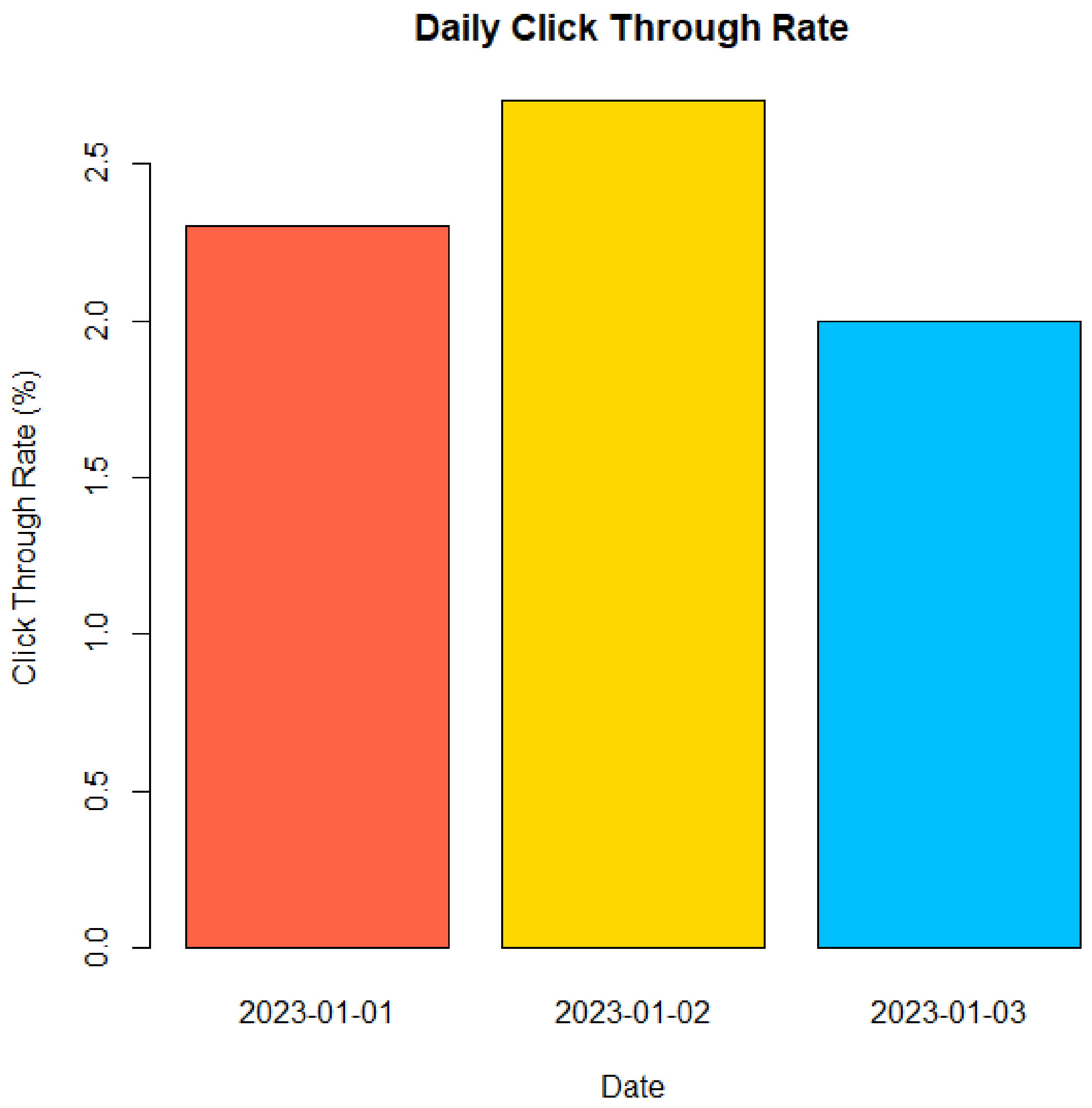
labels=traffic$Date)



2. Generate a scatter plot to explore the relationship between average temperature and population size. Explain any insights.

```
barplot(traffic$CTR,
    names.arg=traffic$Date,
    col=c("tomato","gold","deepskyblue"),
    border="black",
    xlab="Date",
    ylab="Click Through Rate (%)",
```

main="Daily Click Through Rate")



3. Build a table to display the geographic data for each city. Label the table elements.

```
traffic <- data.frame(
  Date=c("2023-01-01","2023-01-02","2023-01-03"),
  PageViews=c(1500,1600,1400),
  CTR=c(2.3,2.7,2.0)
)
traffic$EstimatedClicks <- round(traffic$PageViews * traffic$CTR / 100)
print(traffic)
```

```
> print(traffic)
        Date PageViews CTR EstimatedClicks
1 2023-01-01      1500 2.3              34
2 2023-01-02      1600 2.7              43
3 2023-01-03      1400 2.0              28
> 
```

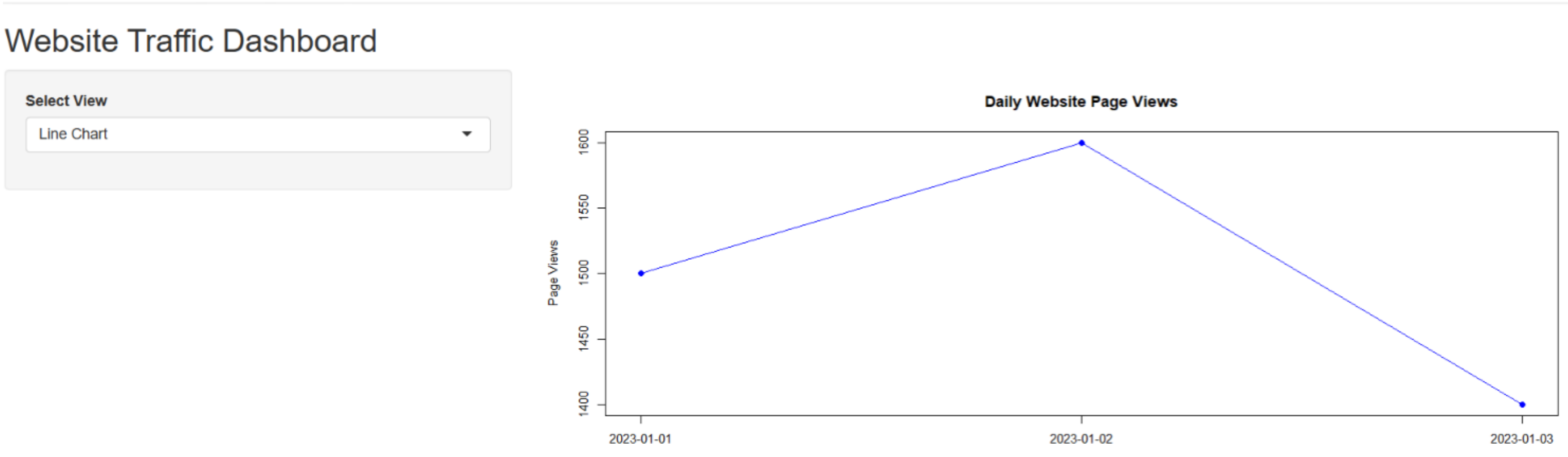
4. Develop a Tableau dashboard combining the map chart, scatter plot, and the table for interactive exploration of geographic data.

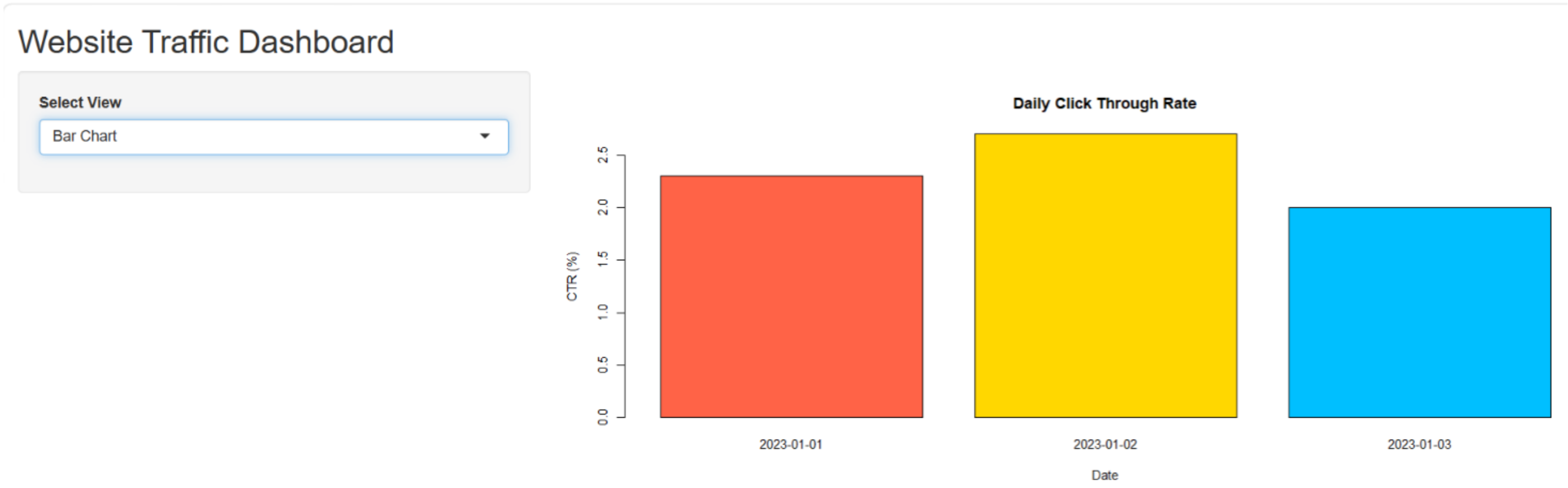
```
library(shiny)

traffic <- data.frame(
  Date=c("2023-01-01","2023-01-02","2023-01-03"),
  PageViews=c(1500,1600,1400),
  CTR=c(2.3,2.7,2.0)
)

ui <- fluidPage(

titlePanel("Website Traffic Dashboard"),

sidebarLayout(

sidebarPanel(

selectInput("view",
      "Select View",
      choices=c("Line Chart",
          "Bar Chart",
          "Traffic Table"))

),
```

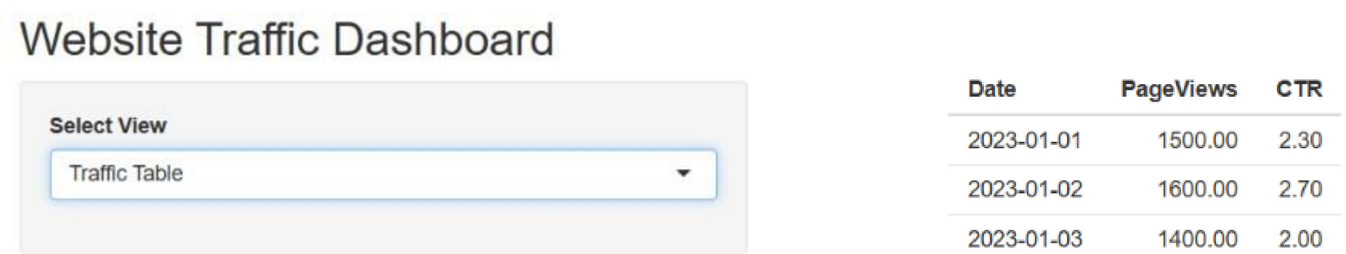
```
mainPanel(

plotOutput("plot"),

tableOutput("table")

)

)

)

server <- function(input,output){

output$plot <- renderPlot({

if(input$view=="Line  Chart"){

plot(traffic$PageViews,
    type="o",
    pch=16,
    col="blue",
    xaxt="n",
    xlab="Date",
    ylab="Page Views",
    main="Daily Website Page Views")
```

```
axis(1,
    at=1:3,
    labels=traffic$Date)

}

else if(input$view=="Bar Chart"){

barplot(traffic$CTR,
       names.arg=traffic$Date,
       col=c("tomato","gold","deepskyblue"),
       border="black",
       xlab="Date",
       ylab="CTR (%)",
       main="Daily Click Through Rate")

}

})

output$table <- renderTable({

if(input$view=="Traffic Table"){

traffic
}
})
}
```

```
shinyApp(ui,server)
```







Date	PageViews	CTR
2023-01-01	1500.00	2.30
2023-01-02	1600.00	2.70
2023-01-03	1400.00	2.00